

# Upper White flood-map completion: interview note

HUC-8 05120201. Methods and claim discipline.  $P(sfha \mid hydro)$  on a 30 m EPSG:5070 grid.

Revisions: 2026-08-27: first interview note (abstract, Table 1, Figure 2, uses, Stage C metrics, limitations). Generated 2026-08-28T02:06:22Z. Git 5d1ed2ecd8a0. Product artifacts at c0e9d5e.

## Abstract

$P(sfha \mid hydro)$  is a map-completion score, not a 1%-annual-chance exceedance probability.

Calibrated OOF PR-AUC (0.36) beats the SFHA-rate baseline (0.10) and a HAND score (0.24).

Five Zone X plants have one wet cell in a 120 m window; site-mean  $P$  is 0.06 to 0.19; none clear 0.50 on the footprint.

THURSDAY POOLS is the only large-inventory terrain hit, and it is neighboring land.

June 2008 Appendix 2 does not cover the industrial core of 05120201.

To three decimals: raw PR-AUC 0.369, calibrated 0.362, Brier 0.073 after isotonic calibration. Stage D samples `p_sfha_calibrated.tif` only. Do not treat 0.75 as a percent chance on the raw raster.

**Table 1. Five Zone X plants ranked on window-max  $P$**

| Plant                 | 2023 on-site release (lb) | Highest $P$ in 120 m | Mean $P$ in 120 m         | What the high cell is        |
|-----------------------|---------------------------|----------------------|---------------------------|------------------------------|
| THURSDAY POOLS        | 257590                    | 0.769                | <code>p_mean</code> 0.152 | neighboring land, unshaded X |
| FGF LLC               | 27335                     | 0.780                | <code>p_mean</code> 0.060 | adjacent hydro (floodway)    |
| ROYAL SPA CORP        | 4950                      | 0.774                | <code>p_mean</code> 0.113 | adjacent hydro (waterbody)   |
| LINDE GAS & EQUIPMENT | 1048                      | 0.763                | <code>p_mean</code> 0.192 | neighboring land, unshaded X |
| MAGNA POWERTRAIN EAST | 0                         | 0.789                | <code>p_mean</code> 0.098 | neighboring land, unshaded X |

Table 1 ranks on the wettest 30 m cell inside a 120 m window. Mean  $P$  is the footprint. At THURSDAY POOLS (`p_mean` 0.152) that cell is neighboring unshaded X, HAND = 0, 120 m from the office point: an edge screen, not a wet footprint. FGF LLC (`p_mean` 0.060) and ROYAL SPA CORP (`p_mean` 0.113) are adjacent-hydro footnotes (floodway corner; waterbody). MAGNA POWERTRAIN EAST (`p_mean` 0.098) is in the list because the rank is window-max  $P$ . Inventory pounds are the table column only.

**Figure 2. Calibrated map and TRI overlay**

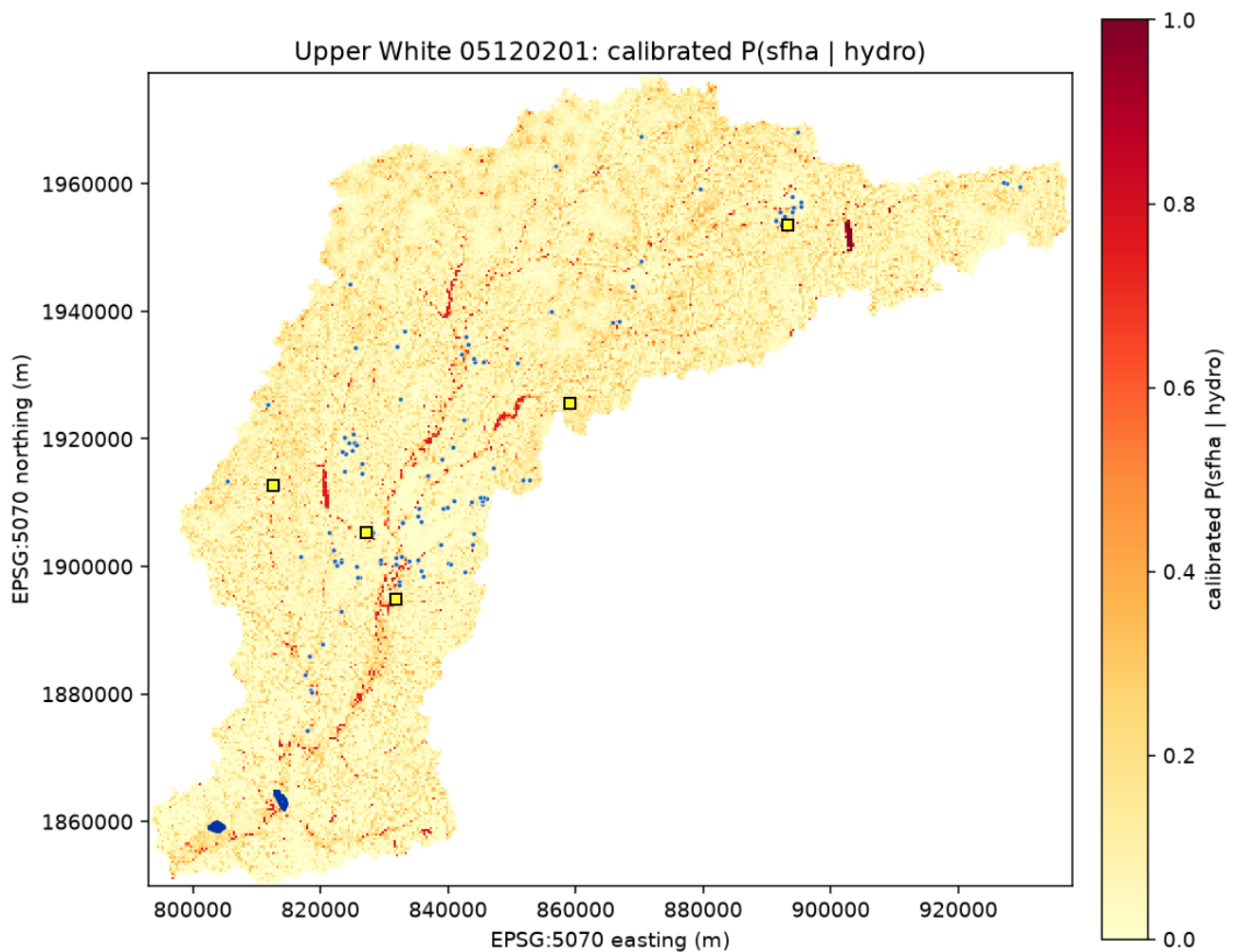


Figure 2. Calibrated  $P(\text{sfha} | \text{hydro})$  for 05120201. Blue dots: 117 in-HUC TRI plants (2023). Red dots: five headline plants. Black line: office point to the window-max cell (yellow square). Blue outlines: June 7-9, 2008 inundation (OFR 2008-1322) code 2, dissolved to Martinsville and Paragon. Interactive copy: [logs/stage\\_d/map.html](logs/stage_d/map.html).

## What you can do with it

### 1. Portfolio (methods and claim discipline).

This is the highest-value use. Geospatial, climate-risk, insurance, and civic-tech readers can see that the work locked a geography (Upper White 05120201), refused a bad label (IndianaMap dropped unshaded Zone X; NFHL layer 28 with where=1=1 restored it), calibrated probabilities instead of waving at 0.75, split window-max from site-mean so five factories were not called flooded, and wrote claim bans into CI. Lead with the five abstract sentences and Figure 2. Show <map.html>, Table 1, and the Stage C metrics below. Do not pitch this as hidden flood zones.

### 2. Interview talk (about 90 seconds).

*I trained a model to complete FEMA floodplain maps in the Upper White basin. It beats a HAND baseline on ranking. When I overlay TRI plants, five Zone X sites have one wet cell in a 120 m window. Site-average probability stays low. The 2008 USGS inundation maps do not cover Indianapolis industry. So this is map completion plus an edge screen, not a plant-level hazard list.*

If they push: calibration, spatial CV, and why Zone X extraction was the actual bug. Those notes are below. They are methods, not a second finding.

### 3. This short note.

Five sentences as abstract, Table 1, Figure 2, one limitations paragraph. That is the sendable object. The repo is martialsystems/indiana\_flood\_completion (private).

## Stage C metrics

| Metric               | Value                                                            |
|----------------------|------------------------------------------------------------------|
| PR-AUC, quoted       | 0.36 (raw 0.369, calibrated 0.362)                               |
| SFHA-rate baseline   | 0.10 (0.097)                                                     |
| HAND score PR-AUC    | 0.24 (0.243)                                                     |
| Brier after isotonic | 0.073                                                            |
| Brier, raw scores    | 0.177 (constant classifier 0.088)                                |
| CV                   | leave-one-HUC-10-out, 17 watersheds, 1-pixel halo                |
| Features             | slope, TWI, HAND, dist_flowline, dist_waterbody, NLCD impervious |
| HSG in model         | no                                                               |
| D sample raster      | p_sfha_calibrated.tif                                            |
| In-HUC TRI 2023      | 117 (101 unshaded X; 16 already mapped hazard)                   |

*Raw booster scores ranked and were miscalibrated (Brier worse than the constant). Isotonic regression on out-of-fold scores, same HUC-10 cuts, wrote the calibrated raster. Rank held (0.369 to 0.362). D never samples raw p\_sfha.tif.*

## If they push

Zone X extract: IndianaMap FIRM 2023 omitted AREA OF MINIMAL FLOOD HAZARD polygons. Monument Circle, a Carmel till-plain cell, and a Delaware field painted as unmapped. D1 is zone\_class == unshaded\_x, so that extract would have dropped Indianapolis mapped Zone X from the overlay. FEMA NFHL MapServer layer 28, where=1=1, is the live FIRM. Interior unshaded X is 6,986,426 cells (89.23%).

Spatial CV: 17 HUC-10 blocks, leave-one-out, no test HUC-10 in train, including a 1-pixel halo. Labels are binary SFHA (floodway included). HAND-nodata cells are excluded and stay nodata on both P rasters.

Max versus mean: buffer-max is one 30 m cell in a 120 m window. Buffer-mean is the footprint. Zero headline rows have mean P at or above 0.50. Adjacent hydro is NHD/FIRM paint: FGF LLC (p\_mean 0.060) floodway corner, ROYAL SPA CORP (p\_mean 0.113) waterbody.

Claim scanner: floodmap.claims fails a report that uses banned tokens, treats P as a 1%-annual-chance exceedance, prints empty D2 without the code-1 / code-2 split, or names a D1 headline plant without p\_mean in the same window. GraphForge pin: whiteforge/.

## Close-out: SHAP

Global SHAP on C features, no HSG: HAND first (mean |SHAP| 1.37), then distance to water, then slope. TWI is last because HAND already ate the wetness signal. At the THURSDAY POOLS max cell (p\_mean 0.152), HAND = 0 is the local driver and the cell is still unshaded X. Write-up, not discovery.

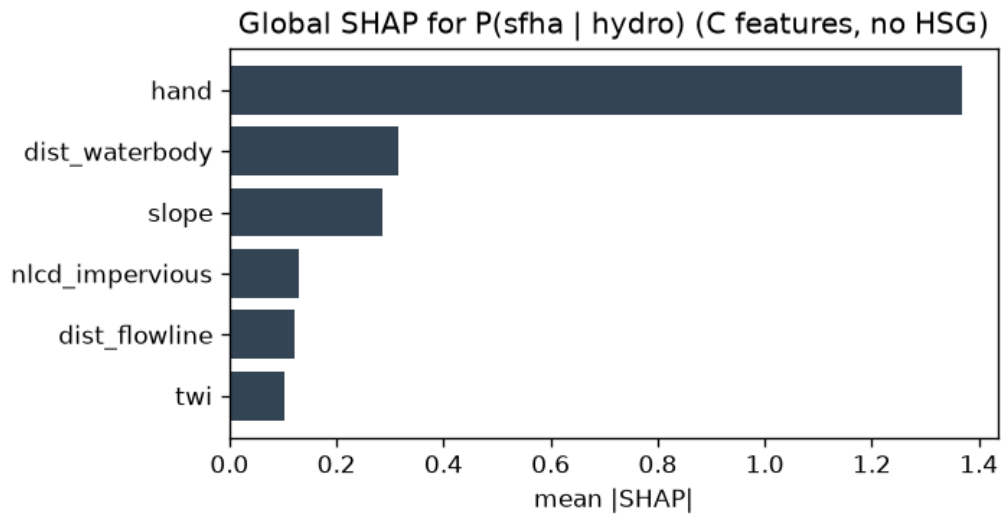


Figure 3. Global SHAP for C features. Close-out, not a new result.

## Limitations

No parcels: adjacent hydro is NHD/FIRM paint, not a cadastral clip; a parcel clip would change certainty on FGF LLC (p\_mean 0.060) and ROYAL SPA CORP (p\_mean 0.113), not the mean-versus-max finding. No soil: hydrologic soil group is not in this model; gSSURGO C2 was not required for the claim graph. Not a FIRM: P does not replace the effective flood map. 2008 is coverage: 117 plants in mask code 1, 0 in code 2; Appendix 2 reaches in the HUC are Martinsville and Paragon only. The map HTML was not opened in a browser for this note; Figure 2 is drawn from the calibrated raster and the D tables.